

2021-Ph-H1-Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

A ball is dropped from rest and hits the ground after 0.40 s.

What is the displacement of the ball during this time?

Solution:

Use the equation:

$$s = \frac{1}{2}gt^2 = \frac{1}{2} \times 9.8 \times (0.40)^2 = 0.78 \text{ m}$$

Answer: B

Guidance for Students:

This is a classic “object dropped from rest” question. Use the equation that relates displacement, time, and acceleration.

Revision Tip:

If you start from rest and accelerate under gravity:

$s = \frac{1}{2}gt^2$ is often quicker than using multiple steps.